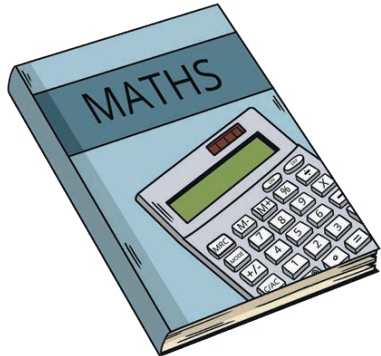


Expanding Brackets Using FOIL



Example

Expand and simplify: $(x + 4)(x + 2)$

Firsts

Outsides

Insides

Lasts

Step One

Multiply the 'Firsts'

$$(x + 4)(x + 2)$$

$$x \times x = x^2$$

Step Two

Multiply the 'Outsides'

$$(x + 4)(x + 2)$$

$$x \times +2 = +2x$$

Step Three

Multiply the 'Insides'

$$(x + 4)(x + 2)$$

$$+4 \times x = +4x$$

Step Four

Multiply the 'Lasts'

$$(x + 4)(x + 2)$$

$$+4 \times +2 = +8$$

Step Five

You have now fully expanded your double brackets. Your next step is to write out the terms you have calculated and collect the like terms.

$$(x + 4)(x + 2)$$
$$x^2 + 2x + 4x + 8$$

Step Six

Simplify your answer:

Your final answer is $x^2 + 6x + 8$

